PROJECT AND TEAM INFORMATION

Project Title

STUDENT ACADEMIC RESULT & COURSE REGISTRATION SYSTEM

Team Name: *Team #*	*CODE RUSHERS*
Team member 1 (Team Lead)	*Gairola, Anirudh - 2510370098* *2510370098@geu.ac.in* 
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PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract

This project is designed to simplify the way colleges manage students' academic records and course registrations. Students often face difficulties in checking their results, selecting courses, and confirming whether they meet the required prerequisites. To address these issues, we aim to develop a system that keeps academic information organized and makes the registration process smoother and more reliable. The system maintains student details, course information, and examination results in a structured format. Data Structures and Algorithms (DSA) are used to make searching and sorting records more efficient, while Object-Oriented Programming (OOP) helps organize the code into separate modules for better readability and maintenance. Before allowing a student to register, the system verifies whether the necessary conditions have been fulfilled. If a course reaches its maximum capacity, the system can also maintain a waiting list for students who still wish to enroll. The project combines programming concepts from C and C++ to develop a practical solution to these common academic management problems. Its main goal is to reduce paperwork and repetitive manual tasks, minimize errors, and make academic record handling and course registration more convenient for students and college staff.

Updated Project Approach and Architecture

We plan to develop our academic management system using a modular approach, in which each feature will be designed separately and later combined to create a complete working system. The C language will be used to implement basic data structures, while C++ will help us apply object-oriented programming concepts, making the code easier to organize, manage, and maintain. The system will include different modules for handling student details, course information, academic results, and course registration. Data structures will be used to store student and course details in an organized manner. Searching and sorting techniques will help retrieve the required information efficiently. Classes and objects in C++ will be used to represent students, courses, and their academic records. During course registration, the system will check whether a student has completed the required prerequisites before allowing them to enroll. If a course has a limited number of seats, a waiting-list facility can be introduced for students who cannot register immediately. The system will also allow academic results to be stored, retrieved, and displayed whenever required. For the initial prototype, we will use local files to store information permanently so that the records remain available even after the program has been closed and run again.

Tasks Completed

Task Completed	Team Member
Finalized the project idea,objective,and system requirements.	ANIRUDH GAIROLA
Planned the system modules for student records and course management.	ANJALI SAIN
Studied and selected suitable DSA concepts for searching and sorting records.	PRATHAM MUKHERJEE
Planned the academic result processing and course registration features.	BISHESH SAHA

Challenges/Roadblocks

One of the main challenges our team may encounter is combining C and C++ concepts while keeping the application properly organized into separate modules. Selecting suitable data structures for storing student records and implementing efficient searching and sorting methods will also require careful planning. Another challenge is making sure that the prerequisite verification works correctly and that the waiting-list system handles courses with limited seats fairly. To deal with these challenges, we plan to divide the work among team members according to their responsibilities and discuss different approaches before implementing them. We will also practise the programming concepts required for the project to improve our understanding. Each module will be developed and tested separately so that errors can be identified and corrected at an early stage. Once the individual modules are working properly, we will integrate them and test the complete application to ensure that all its features work together smoothly.

Tasks Pending

Task Pending	Team Member (to complete the task)
Implement student record management and data structures.	PRATHAM MUKHERJEE
Develop course management and registration modules.	ANIRUDH GAIROLA
Implement searching,sorting,and file-handling features.	ANJALI SAIN
Develop result processing,prerequisite checks,and waiting list functionality.	BISHESH SAHA
Integrate all modules,test the complete system,and prepare the final report and presentation.	ALL MEMBERS

Project Outcome/Deliverables

The main objective of our project is to develop a well-organised academic management system that makes it easier to maintain student records and manage course registrations. The system will provide a structured way to store student information and course details, calculate and display academic results, and handle registration requests. Suitable data structures will be used to keep the records organised and make information retrieval easier.

Another important goal is to implement a proper prerequisite-checking system and manage course availability. This will help students register for courses for which they are eligible and provide a waiting-list facility when all available seats are occupied. File handling will be used to save records permanently so that the information remains available even after the program is closed.

Through this project, we aim to apply C programming, object-oriented concepts in C++, and Data Structures and Algorithms to solve a practical problem in academic management. By the end of the project, we expect to deliver the complete source code, individual program modules, testing documentation, and a final project report.

Progress Overview

The project has currently reached the planning and system design stage. During this phase, our team has finalised the project idea, identified its main objectives, and discussed the requirements needed to develop the Student Academic Result and Course Registration System.

We have also decided how C and C++ will be used in the project. C will help us implement basic data structures, while C++ will be used to apply object-oriented programming concepts. Along with these, we have identified searching, sorting, file handling, prerequisite verification, and waiting-list management as the major features of the system.

At present, our team is focusing on planning the implementation process and selecting suitable programming techniques for each feature. We are also working on dividing the system into smaller modules so that development and testing can be carried out in an organised manner.

In the next stage, we plan to integrate the individual modules, handle possible errors, and test the complete system to check whether all the features work correctly together.

Codebase Information

The project will include C and C++ source files divided into separate modules for managing student records, course details, academic results, and course registration. To keep the code organised and make teamwork easier, we plan to maintain the project on GitHub. It will help us track changes, manage different versions of the code, and work together more efficiently. We will regularly update the repository as the project progresses, including the initial setup, development of individual modules, integration of features, correction of errors, and final testing of the complete system.

Repository Link: https://github.com/sen339853-rgb/STUDENT-ACADEMIC-RESULT-COURSE-REGISTRATION-SYSTE.git

Branch: Main

Key Commits: Initial setup, module development, and feature integration.

Testing and Validation Status

Test Type	Status (Pass/Fail)	Notes
Student Record Management	Pass	Testing record entry & retrieval
Course Management	Pass	Verify course details & availability
Searching and Sorting	Pass	Check record organization & retrieval
Prerequisite verification	Pending	Validate course eligibility
File Handling	Pending	Test saving & loading records
System Integration	Pending	Test all modules together

Deliverables Progress

The project requirements and overall system design have been finalised. Our team has also selected C and C++ as the programming languages and identified the Data Structures and Algorithms (DSA) and Object-Oriented Programming (OOP) concepts required for development.

The system is planned to include four main modules: student record management, course management, academic result processing, and course registration. The development and integration of these modules are currently in progress or planned for the upcoming phase. Additional features, such as prerequisite verification, waiting-list management, and file handling, will also be implemented to improve the system's functionality. Once the modules are developed, we will test the application and debug any errors found during the process. The final deliverables will include the complete source code of the tested academic management system, a detailed project report, and a presentation covering the system design, major features, implementation, and results.